An Open-Access Deep Learning Tool for Early Detection of Chiari Type I Malformation via MRI Classification

Anonymous Authors[1]

Abstract

Chiari Type I Malformation (CM-1) is a congenital neurological condition in which the cerebellar tonsils descend $\geq$5 mm below the foramen magnum, compressing the brainstem and disrupting cerebrospinal fluid dynamics. In Colombia and much of Latin America, the scarcity of specialized neuroradiology expertise outside major urban centers leads to diagnostic delays that can span years, limiting access to timely surgical intervention. We present a complete, reproducible pipeline—from open-access dataset construction to public web deployment—for automated CM-1 detection in mid-sagittal T1-weighted brain MRI. Our system applies two-phase transfer learning on DenseNet121, EfficientNetB0, and ResNet50V2, trained on a curated dataset of 156 real clinical images (126 CM-1, 30 normal) obtained from Radiopaedia.org under a CC BY-NC-SA 4.0 license and preprocessed with CLAHE contrast enhancement. Evaluated on an independent external test set of 131 images, DenseNet121 achieved AUC-ROC 0.943, sensitivity 0.852, specificity 0.933, and F1-score 0.910. Gradient-weighted Class Activation Mapping (Grad-CAM) confirms that the model attends to the posterior cranial fossa and foramen magnum—the anatomical structures that define CM-1 radiologically. The tool is publicly deployed, requires no registration, runs inference in under one second, and implements privacy-by-design by never writing images to disk. To the best of our knowledge, this is the first freely accessible deep-learning diagnostic aid for CM-1 developed in and for a Latin American context.

[1]Anonymous Institution, Anonymous City, Anonymous Region, Anonymous Country. Correspondence to: Anonymous Author <anon.email@domain.com>.

Preliminary work. Under review by the International Conference on Machine Learning (ICML). Do not distribute.

1. Introduction

Rare and orphan diseases (RODs) represent a critical challenge for health systems worldwide. In May 2025, the World Health Organization approved resolution EB156/15, recognizing that more than 300 million people suffer from one of over 7,000 known rare diseases, most of which begin in childhood and produce severe physical, emotional, and economic burdens (World Health Organization, 2025). In Latin America and the Caribbean, access to innovative treatments for these conditions remains deeply unequal: structural analyses of the region show that regulatory approval does not translate into effective financial coverage, and only countries with the most robust health systems achieve sustained access (Sugg Herrera et al., 2026).

Colombia was the first Latin American country to enact legislation in favor of ROD patients via Law 1,392 of 2010, recognizing them as a matter of special public health interest. Yet implementation has been partial: of 2,247 diseases in the national registry, only 29 are under active clinical study, concentrated in Bogotá, Medellín, and Cali. The scarcity of specialized personnel severely limits diagnostic access for more than half of Colombian territory (Roldan et al., 2026; Leal-Bernal et al., 2026).

Chiari Type I Malformation (CM-1) exemplifies these consequences with particular severity. It is a congenital neurological condition characterized by the descent of the cerebellar tonsils below the foramen magnum, compressing the brainstem and altering cerebrospinal fluid (CSF) circulation (Avellaneda Fernández et al., 2009). Its ophthalmological manifestations—papilledema, nystagmus, and diplopia—affect 53% of pediatric cases and can serve as early diagnostic indicators (Dhawan et al., 2026). At the cognitive level, deficits in processing speed, visuospatial ability, and executive function have been documented in adults, with microstructural anomalies in the cerebellar peduncles correlating with worse attention and greater pain (Fouda et al., 2026; Houston et al., 2026). The diagnostic gold standard is mid-sagittal T1-weighted MRI, where the extent of tonsillar herniation is measured relative to the foramen magnum; a descent $\geq$5 mm is considered diagnostic, though lesser descents with syringomyelia or compatible symptoms are also clinically significant (Visocchi et al., 2025). Treatment of choice in symptomatic patients is posterior fossa decompression with duroplasty, and the WFNS Delphi consensus recommends surgery also in asymptomatic patients with MRI-documented progression (Visocchi et al., 2025).

Artificial intelligence, and computer vision in particular, offers promising solutions for this diagnostic gap. A review published in *JAMA Neurology* documented that computer vision models can measure disease severity and characterize therapeutic outcomes with high precision, outperforming traditional clinical instruments (Friedrich et al., 2025). For small labeled datasets—an inherent characteristic of orphan diseases—transfer learning on pretrained CNN architectures has demonstrated outstanding performance in medical image classification, enabling faster and more reproducible diagnoses while mitigating overfitting risk (Sharma et al., 2025; Mallik et al., 2026). At a systemic scale, tools such as RareAlert, which integrates reasoning from ten large language models calibrated via machine learning, achieved AUC 0.917 on an independent test set covering more than 7,000 rare conditions, demonstrating that rare disease detection can be reconceptualized as a universal uncertainty-resolution process (Chen et al., 2026).

Prior work on deep learning for CM-1 detection includes Tanaka et al. (2022), who trained CNNs on hospital-grade MRI data and reported AUC values above 0.90, and Mesin et al. (2022), who demonstrated machine learning's value in supporting treatment decisions for CM-1. Urbizu et al. (2017) showed earlier that cephalometric MRI measurements could reliably predict CM-1, identifying the basion as a key morphometric indicator. However, all prior studies relied on datasets acquired under controlled clinical conditions with proprietary access, and none resulted in a publicly accessible diagnostic tool.

We address these gaps with three contributions: (1) a reproducible pipeline for building a CM-1 MRI dataset from openly licensed sources (CC BY-NC-SA 4.0, Radiopaedia.org) that enables community extension; (2) a rigorous two-stage evaluation combining stratified K-Fold cross-validation with an independent external test set, comparing DenseNet121 (Huang et al., 2017), EfficientNetB0 (Tan & Le, 2019), and ResNet50V2 (He et al., 2016) under identical conditions; and (3) a freely accessible web application requiring no registration, running inference in under one second, and implementing privacy-by-design—the first such tool for CM-1 in a Latin American context.

2. Dataset and Preprocessing

2.1. Dataset Construction

All images were sourced from Radiopaedia.org under a CC BY-NC-SA 4.0 license, which explicitly permits academic research and publication (Creative Commons, 2024). The raw dataset contains 156 mid-sagittal T1-weighted MRI images: 126 CM-1 positive and 30 normal brain scans. Positive cases span a wide age range (1–70 years, mean 32.3; 47 F, 19 M, 10 unknown) and include diverse presentations: isolated tonsillar herniation, herniation with syrinx, herniation with hydrocephalus, and acquired CM-1. Normal cases cover ages 0.5–55 years (11 F, 12 M, 2 unknown) and include studies performed for diverse non-structural indications.

An automated quality-control pass verified all 156 images (100% yield) against three criteria: maximum pixel value >10 (rejects blank/black frames), standard deviation >5 (rejects zero-information scans), and minimum dimension ≥ 32 px. Raw images arrived in three color modes (L, RGB, RGBA) and multiple resolutions (most common: 512×512 px).

2.2. Manual Posterior Fossa Cropping

The anatomical region of interest for CM-1 classification is the posterior cranial fossa: the skull compartment housing the cerebellum, brainstem, and foramen magnum. Including the full MRI frame in training would expose the model to irrelevant information and risk learning spurious correlations unrelated to tonsillar descent. Because the dataset contains images from multiple scanners and acquisition protocols with heterogeneous fields of view and angulation, a fixed programmatic crop produced inconsistent results. Therefore, each image was manually inspected and cropped to include the cerebellar tonsils, the foramen magnum, and the entire brainstem.

2.3. Preprocessing Pipeline

A single deterministic preprocessing function was applied identically during both training and inference, guaranteeing that the tensor received by the model at deployment time is statistically equivalent to those seen during training. This consistency is particularly critical in medical imaging pipelines, where any divergence between training and serving conditions can silently degrade model performance even when the model weights are correct. The pipeline executes four sequential steps.

Grayscale conversion. Because the dataset contains images in mixed color modes—single-channel grayscale, three-channel RGB, and four-channel RGBA—the first step converts every image to a uniform single luminance channel. This eliminates format heterogeneity without any loss of diagnostic content, since T1-weighted brain MRI encodes all clinically relevant contrast in luminance intensity rather than color.

Local contrast enhancement. Contrast Limited Adaptive Histogram Equalization (CLAHE) is applied over a grid of

8 × 8 pixel tiles with a clip limit of 2.0. Unlike global histogram equalization, which redistributes intensities across the entire image and can flatten diagnostically relevant gradients, CLAHE operates locally and caps its amplification factor to prevent noise over-enhancement. In posterior fossa MRI, this step substantially improves the visibility of the boundary between cerebellar tissue and cerebrospinal fluid—the contrast gradient that encodes tonsillar position and is therefore central to CM-1 classification.

Resizing and intensity normalization. Each image is downsampled to 224×224 pixels using Lanczos interpolation, which applies a high-quality sinc-windowed kernel that preserves anatomical edges better than simpler resampling methods. Proportional scaling is maintained and any remaining space is filled with symmetric zero-padding, preserving anatomical aspect ratios without geometric distortion. Pixel intensities are then linearly mapped to the range $[0, 1]$, matching the normalization convention used when the architectures were pretrained on ImageNet.

Channel replication and batch formatting. ImageNet-pretrained networks expect three-channel inputs. To satisfy this requirement while preserving the single-channel nature of grayscale MRI, the luminance channel is replicated across all three color channels, producing a $224 \times 224 \times 3$ tensor in which all channels carry identical information. A batch dimension is prepended to yield the final input of shape $1 \times 224 \times 224 \times 3$, which is the exact input signature declared for all three architectures.

Training augmentation (applied only during model training) consisted of horizontal flip ($p = 0.5$), brightness jitter $\pm 10\%$ ($p = 0.5$), and additive Gaussian noise ($\sigma = 0.008$, $p = 0.4$).

Figure 1 illustrates the effect of CLAHE on a representative CM-1 case. Before processing, the intensity histogram is concentrated in mid-range values, producing low contrast in the posterior fossa. After CLAHE, the histogram redistributes toward both extremes of the dynamic range, improving anatomical definition without saturating high-intensity regions or amplifying background noise.

2.4. Dataset Partitioning and Class Imbalance

The raw dataset is imbalanced at a 4.2:1 positive-to-negative ratio. We adopted *undersampling* of the majority class rather than synthetic oversampling (SMOTE): interpolation between MRI images introduces visually non-realistic features that corrupt the CNN feature space, whereas using only real clinical images ensures the model learns genuine anatomical patterns.

The cross-validation training partition contained 60 balanced images (30 CM-1, 30 normal, `SEED=42`). An independent external evaluation set of 131 images (101 CM-1, 30 normal) was held out entirely from all model selection, hyperparameter tuning, and training decisions. Table 1 summarizes the final dataset composition.

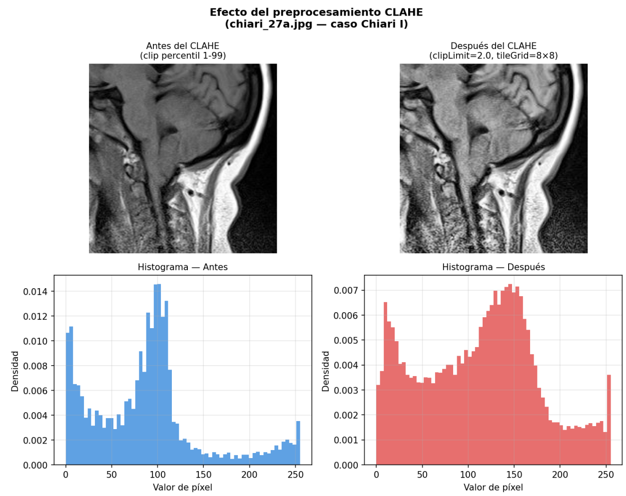

Figure 1. Effect of the CLAHE preprocessing step on a CM-1 mid-sagittal T1-weighted MRI (case `chiari_27a.jpg`). *Left column:* image after percentile 1–99 clipping and its intensity histogram. *Right column:* image after CLAHE (`clipLimit=2.0`, `tileGridSize=8×8`) and its redistributed histogram. The posterior fossa shows improved tissue-CSF contrast without noise amplification.

Table 1. Final dataset composition and partitioning.

Partition	CM-1	Normal	Total
Training (K-Fold)	30	30	60
External eval	101	30	131
Raw dataset	126	30	156

3. Model Architecture and Training

3.1. Architecture Comparison

Three ImageNet-pretrained architectures were compared under identical conditions (Table 2). All share the same classification head appended over the convolutional base: GlobalAveragePooling → BatchNormalization → Dense(32, ReLU, L2=10^{-3}) → Dropout(0.5) → Dense(1, sigmoid). DenseNet121 (Huang et al., 2017) features dense connectivity in which each layer receives feature maps from all preceding layers, promoting feature reuse and parameter efficiency. With only 7.07 M total parameters and 34,881 trainable parameters in Phase 1, it is well matched to the low-data regime. Grad-CAM maps are extracted from its final dense block concatenation layer. EfficientNetB0 (Tan & Le, 2019) optimizes depth, width, and resolution via a compound scaling factor (4.10 M total parameters); its archi-

Table 2. Architecture comparison. Trainable params in Phase 1 (frozen base). $\star$ = selected for production.

Architecture	Total params	Trainable (Ph1)
DenseNet121 $\star$	7,074,433	34,881
EfficientNetB0	4,095,716	43,585
ResNet50V2	23,638,593	69,697

tecture is designed for large-scale datasets, and Grad-CAM is extracted from its top convolutional layer. ResNet50V2 (He et al., 2016), with pre-activated residual connections (23.64 M total parameters), imposes the heaviest parameterization relative to available training data; its post-activation layer serves as the Grad-CAM target.

3.2. Two-Phase Transfer Learning

Phase 1 (feature extraction): the convolutional base is frozen and only the classification head is trained, allowing the network to learn a task-specific decision boundary on top of general-purpose ImageNet features without modifying them. Training runs for up to 40 epochs with a learning rate of 10^{-3}, batch size of 8, and binary cross-entropy loss optimized via Adam.

Phase 2 (fine-tuning): the top 10 layers of the convolutional base are unfrozen and trained jointly with the classification head at a substantially reduced learning rate of 10^{-6}, for up to 30 additional epochs. This allows the deepest representations to adapt gradually to the posterior fossa domain without overwriting the low- and mid-level features inherited from ImageNet pretraining.

Both phases employ early stopping monitored on validation AUC with a patience of 15 epochs, and an adaptive learning rate schedule that halves the rate after 8 epochs without improvement, down to a minimum of 10^{-8}.

All experiments use stratified 5-fold cross-validation with shuffling and a fixed random seed, yielding approximately 48 training and 12 validation images per fold while preserving the class ratio in every split. Model weights are re-initialized from scratch at the start of each fold to prevent any information leakage across folds. The checkpoint with the highest validation AUC within each fold is retained, and the single best checkpoint across all five folds is used for external evaluation. The full pipeline runs on a consumer laptop (Intel i5-10300H, 16 GB RAM, 4 GB GPU VRAM), deliberately chosen to ensure that the results are reproducible without access to specialized computing infrastructure.

3.3. Grad-CAM Interpretability

Gradient-weighted Class Activation Maps (Selvaraju et al., 2017) were computed via a two-output auxiliary model: activation maps of the final convolutional layer and the sigmoid output probability. Using `tf.GradientTape`, gradients of the CM-1 class probability with respect to the activation maps are computed, spatially averaged to produce per-channel weights (*pooled gradients*), and used to weight each channel. The resulting weighted sum is ReLU-activated and normalized to $[0, 1]$; a `jet` colormap (blue = low, red/yellow = high activation) is overlaid on the original image. Grad-CAM provides post-hoc evidence that the model's decisions are grounded in clinically relevant anatomy rather than imaging artifacts.

Table 3. 5-Fold cross-validation results (mean $\pm$ std, $n = 60$ images).

Metric	DenseNet121	EfficientNetB0	ResNet50V2
AUC-ROC	0.506 ± 0.156	0.478 ± 0.137	0.456 ± 0.272
F1	0.546 ± 0.206	0.120 ± 0.268	0.486 ± 0.173
Accuracy	0.517 ± 0.109	0.533 ± 0.075	0.483 ± 0.160
Sensitivity	0.667 ± 0.333	0.100 ± 0.224	0.500 ± 0.200

4. Results

4.1. Cross-Validation Performance

Table 3 reports mean $\pm$ standard deviation across 5 folds. All three architectures exhibit high variance, a statistically expected consequence of the fold size: with 12 validation images per fold, a single misclassification shifts AUC by $>$0.10. High standard deviations are direct evidence of estimator instability at this scale, not evidence of poor learning.

Per-architecture behavior. DenseNet121 showed the most balanced behavior: the only architecture with mean sensitivity above 0.5, indicating it detected CM-1 cases in most folds rather than collapsing toward the majority class. EfficientNetB0 exhibited a degenerate pattern: specificity 0.967 with sensitivity 0.100 means the model classified nearly everything as Normal, maximizing specificity at the cost of near-zero recall. This reflects the mismatch between EfficientNet's architecture—optimized for large-scale datasets—and the available 60 training images. ResNet50V2 showed the highest AUC variance ($\pm$0.272), with 23.6 M parameters far exceeding what the training set can support.

Figure 2 visualizes the metric distributions across all five folds. The near-complete whisker ranges in AUC and sensitivity confirm high estimator variance at micro-fold scale. EfficientNetB0's degenerate behavior is immediately visible: median specificity $\approx$1.0 versus near-zero sensitivity across all folds. DenseNet121's boxes are the most centered and least dispersed in all four panels.

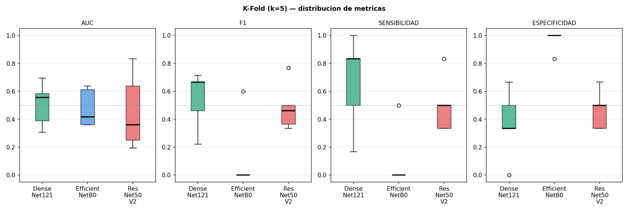

Figure 2. Distribution of cross-validation metrics across 5 folds for DenseNet121 (green), EfficientNetB0 (blue), and ResNet50V2 (red). The dashed reference line marks the random-classifier baseline (0.5). Wide whiskers reflect the high variance of AUC estimation at only 12 validation images per fold. EfficientNetB0's collapsed sensitivity ($\approx$0) with near-perfect specificity ($\approx$1) confirms majority-class prediction.

Table 4. External evaluation — DenseNet121 ($n = 131$, threshold = 0.5).

Metric	Value	Clinical interpretation
AUC-ROC	0.9426	High discriminative capacity
Accuracy	0.8702	114/131 correct
Sensitivity	0.8515	86/101 CM-1 detected
Specificity	0.9333	28/30 normals correct
F1-Score	0.9101	Precision–recall balance
TP / TN	86 / 28	
FP / FN	2 / 15	

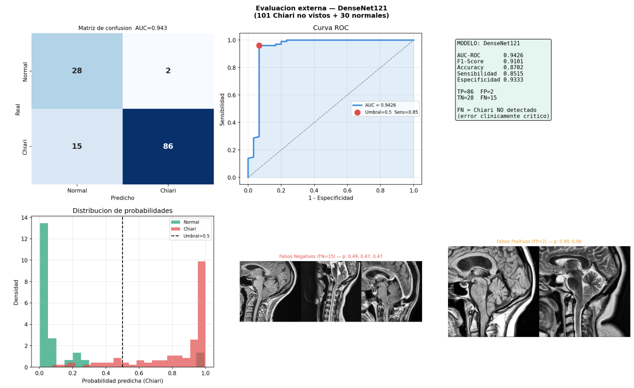

Figure 3. External evaluation dashboard — DenseNet121 on 131 unseen images (101 CM-1, 30 normal). *Top left:* confusion matrix (AUC = 0.943). *Top center:* ROC curve; the red dot marks the operating point at threshold 0.5 (sensitivity 0.85). *Top right:* summary metrics. *Bottom left:* predicted probability distributions for each class; normal cases concentrate near 0, CM-1 cases near 1. *Bottom center:* three representative false negatives (FN, $p \approx 0.47$–0.49) showing subtle or atypical tonsillar descent. *Bottom right:* two false positives (FP, $p \approx 0.98$–0.99) showing posterior fossa morphology at the boundary of normal variation.

4.2. External Evaluation — DenseNet121

The best DenseNet121 checkpoint was evaluated on the 131-image external set (101 CM-1, 30 normal), unseen during all training and model selection. Table 4 presents the full results. The confusion matrix records VP = 86, VN = 28, FP = 2, FN = 15.

Figure 3 presents the complete external evaluation dashboard: confusion matrix, ROC curve, probability distribution, and representative false-negative and false-positive cases.

K-Fold vs. external discrepancy. The gap between K-Fold AUC (0.506) and external AUC (0.943) reflects estimator variance at micro-fold scale, not overfitting. The external set—11$\times$ larger than any single validation fold—provides a reliable low-variance estimate of true generalization. This interpretation is consistent with the theoretical behavior of AUC estimators on small samples and is supported by the high standard deviations visible in Table 3.

Clinical relevance of error types. In CM-1 screening, sensitivity (recall of true CM-1 cases) is the clinically critical metric: a false negative—failing to alert a physician to an existing malformation—carries a far greater cost than a false positive that triggers specialist follow-up. Our model achieves 0.933 specificity (only 2 false alarms across 30 normal cases) while its 0.852 sensitivity correctly identifies 86 of 101 CM-1 cases. The 92.7 percentage-point spread between the two documented test probabilities (6.0% for a normal case, 98.7% for a CM-1 case) illustrates the model's high discriminative confidence at the extremes of the probability spectrum.

4.3. Grad-CAM Validation

Grad-CAM activation maps were computed on the final convolutional layer of DenseNet121 for all external cases. **Correctly classified CM-1 cases:** heat consistently concentrates in the posterior cranial fossa, specifically at the cerebellar tonsils, the foramen magnum, and the inferior brainstem—the exact anatomical structures defining CM-1 radiologically. **Correctly classified normal cases:** activation is diffuse and of low magnitude across the entire posterior fossa, consistent with the absence of tonsillar herniation. **False positive cases:** activation overlaps the posterior fossa but with lower peak intensity and greater spatial dispersion than in true positives—consistent with a model detecting ambiguous morphology rather than spurious correlations. These patterns confirm that the model has learned genuine anatomical discriminators, not scanner-specific artifacts or acquisition protocol signatures.

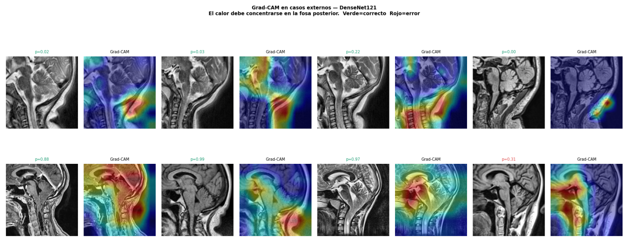

Figure 4. Grad-CAM activation maps on external cases — DenseNet121 (`conv5_block16_concat` layer). Each pair shows the original MRI (left) and the overlaid heat map (right; blue = low, red/yellow = high activation). *Top row (normal cases, p=0.02–0.22):* activation is diffuse and low-magnitude across the posterior fossa, consistent with absent tonsillar herniation. *Bottom row (CM-1 cases, p=0.88–0.99):* heat consistently concentrates at the cerebellar tonsils, foramen magnum, and inferior brainstem — the exact anatomical structures defining CM-1. The rightmost bottom case (p=0.31, false negative) shows diffuse, low-intensity activation, explaining the model's uncertainty. Green probability labels = correct classification; red = error.

5. Web Application

The trained DenseNet121 model was exported to an open interoperable format using the ONNX standard, reducing the deployment artifact from approximately 500 MB to approximately 27 MB and enabling inference without any dependency on the original training framework in the production environment.

The application is built with Streamlit (Streamlit Inc., 2024) and follows a *privacy-by-design* architecture across three principles. First, uploaded images are held exclusively in server memory and are never written to disk at any point in the processing pipeline. Second, the inference session is instantiated once per server lifecycle (approximately 0.3 s cold start) and reused across all subsequent requests, eliminating per-prediction loading latency. Third, no authentication, registration, or session tracking of any kind is required from the user.

The prediction flow is fully sequential: the user uploads a mid-sagittal T1-weighted MRI image, which is passed through the same preprocessing pipeline used during training and then fed to the ONNX inference engine, which returns a scalar probability in $[0, 1]$. If the probability meets or exceeds the decision threshold of 0.5, the interface displays a CM-1 alert; otherwise a normal result is shown. In both cases the numeric probability, a proportional progress bar, and an explicit academic disclaimer are displayed, communicating that the tool is a supplementary aid and does not replace specialist medical judgment—a necessary transparency measure given the model's known false-negative rate of 14.9%.

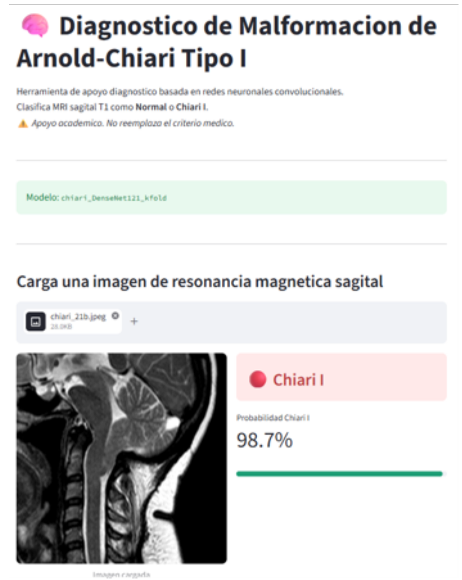

Figure 5. Web application interface showing a CM-1 prediction. The model assigns a probability of 98.7% to the uploaded mid-sagittal T1-weighted MRI, correctly classifying it as Chiari I. The academic disclaimer is displayed in all cases, regardless of the predicted class.

The application is publicly available at an anonymized URL [removed for blind review]. Source code is available at an anonymized repository [removed for blind review].

Figure 5 shows the application interface processing a confirmed CM-1 case, where the model assigns a probability of 98.7%, well above the decision threshold of 0.5. The academic disclaimer is permanently visible regardless of the prediction outcome.

6. Discussion

Strengths. The most relevant result is the 0.852 sensitivity on 131 external images, meaning the model detected 86 of 101 CM-1 cases it had never seen. In the context of a diagnostic aid for an orphan disease with a structural multi-year delay in Colombia, an 85% detection rate on external cases is clinically significant. Grad-CAM validation adds a dimension of confidence that numbers alone cannot provide: the model not only achieves high AUC but does so for the anatomically correct reasons. The entire system runs on a personal computer with a consumer-grade GPU, demonstrating that high-performance medical AI is achievable without specialized compute infrastructure—a key consideration for resource-limited Latin American research groups.

Limitations. Three limitations must be explicitly acknowledged. First, the training set of 60 images is small—a direct consequence of working with an orphan disease and openly licensed data. Transfer learning from ImageNet partially mitigates this, but high K-Fold variance confirms residual statistical instability. Second, all images originate from a single source (Radiopaedia.org) with a demographic and equipment distribution biased toward certain countries; Colombian or more broadly Latin American patients are not explicitly represented, and the model may underperform on lower-resolution equipment or non-standard acquisition protocols. Third, the tool has not undergone prospective clinical validation with licensed neuroimaging specialists on institutional Colombian cases; such validation is a necessary step before any deployment beyond research use.

7. Conclusion

We presented a complete, reproducible pipeline for early detection of Chiari Type I Malformation in mid-sagittal brain MRI, from open-access dataset construction through two-phase transfer learning to a public web deployment. DenseNet121 achieved AUC 0.943, sensitivity 0.852, and specificity 0.933 on an independent external test set of 131 images, with Grad-CAM confirming anatomically coherent decision making. The resulting tool is publicly available, processes images in under one second, and implements privacy-by-design, representing the first open-access diagnostic aid for CM-1 in Latin America. It provides a reproducible baseline and a deployed system upon which future work can build toward clinically validated, community-extended rare disease detection in low-resource settings.

Impact Statement

This work develops a publicly accessible tool intended to reduce diagnostic delays for a rare neurological condition that disproportionately affects patients in Latin America with limited access to specialized neuroradiology. The tool outputs a probability score alongside a mandatory disclaimer that it is not a clinical diagnostic instrument and does not replace specialist judgment; it is explicitly designed to support, not substitute, medical decision making. Potential risks include over-reliance by untrained users; these are mitigated through the on-screen disclaimer, the published false-negative rate of 14.9%, and the open publication of all known limitations. All training images were obtained under explicit open-access licenses (CC BY-NC-SA 4.0) from Radiopaedia.org; no personal data from application users is collected, stored, or transmitted.

References

Avellaneda Fernández, A., Isla Guerrero, A., Izquierdo Martínez, M., et al. Malformations of the craniocervical junction (Chiari type I and syringomyelia): classification, diagnosis and treatment. *BMC Musculoskeletal Disorders*, 10(Suppl 1):S1, 2009. doi: 10.1186/1471-2474-10-S1-S1.

Chen, X., Zhou, H., Yi, H., et al. RareAlert: Aligning heterogeneous large language model reasoning for early rare disease risk screening. *arXiv*, 2026. doi: 10.48550/arXiv.2601.18132.

Creative Commons. Attribution-NonCommercial-ShareAlike 4.0 international (CC BY-NC-SA 4.0), 2024. URL https://creativecommons.org/licenses/by-nc-sa/4.0/deed.es.

Dhawan, A., Dhawan, J., Sharma, A. N., Azzam, D. B., Fehlings, M. G., Quinn, M. P., and Albreiki, D. Ophthalmic manifestations of Chiari malformation: a systematic review. *Canadian Journal of Ophthalmology*, 2026. doi: 10.1016/j.jcjo.2026.03.026.

Fouda, M. A., Srhir, A., Zakursky, E., et al. Hidden impacts, visible deficits: a comprehensive systematic literature review of the cognitive impairment in patients with Chiari malformation type I. *Neurosurgery Review*, 49:45, 2026. doi: 10.1007/s10143-025-03967-3.

Friedrich, M. U., Relton, S., Wong, D., and Alty, J. Computer vision in clinical neurology. *JAMA Neurology*, 82 (4):407–415, 2025. doi: 10.1001/jamaneurol.2024.5326.

He, K., Zhang, X., Ren, S., and Sun, J. Identity mappings in deep residual networks. In *European Conference on Computer Vision (ECCV)*, pp. 630–645, 2016. doi: 10.1007/978-3-319-46493-0_38.

Houston, J. R. et al. High-resolution diffusion tensor imaging in Chiari malformation. *Cerebellum*, 25:45, 2026. doi: 10.1007/s12311-026-01991-7.

Huang, G., Liu, Z., Van der Maaten, L., and Weinberger, K. Q. Densely connected convolutional networks. In *Proceedings of the IEEE/CVF CVPR*, pp. 4700–4708, 2017. doi: 10.1109/CVPR.2017.243.

Leal-Bernal, C. et al. Current landscape of orphan disease clinical research in Colombia. *Journal of Rare Diseases*, 5:28, 2026. doi: 10.1007/s44162-026-00161-x.

Mallik, S. et al. *Computer Vision in Healthcare*. CRC Press, 2026. doi: 10.1201/9781003541011.

Mesin, L., Ponzio, F., Carlino, C. F., et al. A machine learning approach to support treatment identification for Chiari I malformation. *Applied Sciences*, 12(18):9039, 2022. doi: 10.3390/app12189039.

Roldan, E. J. A. et al. Meeting abstracts icord, 2026.

Selvaraju, R. R., Cogswell, M., Das, A., Vedantam, R., Parikh, D., and Batra, D. Grad-CAM: Visual explanations from deep networks via gradient-based localization. In

Proceedings of the IEEE ICCV, pp. 618–626, 2017. doi: 10.1109/ICCV.2017.74.

Sharma, R., Kumar, N., and Sharma, V. Computer vision for disease detection. In *Computer Vision in Healthcare*. CRC Press, 2025. doi: 10.1002/9781394278695.ch4.

Streamlit Inc. Streamlit documentation, 2024. URL https://docs.streamlit.io.

Sugg Herrera, D. et al. Structural enablers of rare disease treatment coverage in latin america. *Journal of Market Access & Health Policy*, 14(1):13, 2026. doi: 10.3390/jmahp14010013.

Tan, M. and Le, Q. V. EfficientNet: Rethinking model scaling for convolutional neural networks. In *Proceedings of the 36th ICML*, pp. 6105–6114, 2019.

Tanaka, K. W., Russo, C., Liu, S., Stoodley, M. A., and Di Ieva, A. Use of deep learning in the MRI diagnosis of Chiari malformation type I. *Neuroradiology*, 64(8): 1585–1592, 2022. doi: 10.1007/s00234-022-02921-0.

Urbizu, A., Martin, B. A., Moncho, D., Rovira, A., Poca, M. A., Sahuquillo, J., Macaya, A., and Español, M. I. Machine learning applied to neuroimaging for diagnosis of adult classic Chiari malformation: role of the basion as a key morphometric indicator. *Journal of Neurosurgery*, 129(3):779–791, 2017. doi: 10.3171/2017.3.JNS162479.

Visocchi, M. et al. Indications for surgery in Chiari malformation: WFNS recommendations. *Spine*, 50(11):760–766, 2025. doi: 10.1097/BRS.0000000000005288.

World Health Organization. Seventy-eighth World Health Assembly — daily update, 24 May 2025. WHO Official Website, 2025.